

A Collaborative Low-Code Platform for Quantum Application Design and Execution



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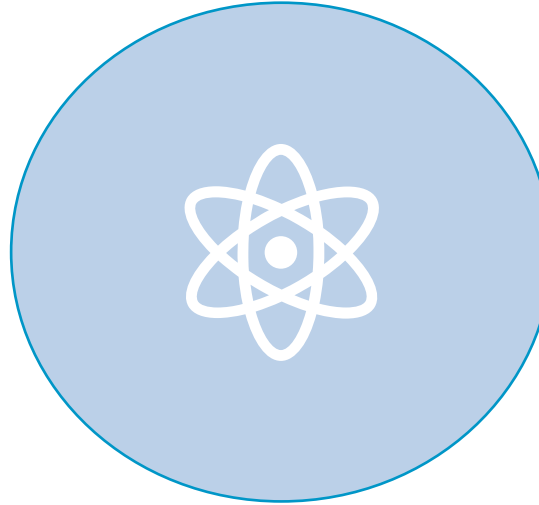
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Institute of Architecture of Application Systems

Quantum Computing in Industry

Motivation

Quantum computing enables **greater precision, faster processing, and improved energy efficiency** for certain problems ^[1,2,3]



Global public funding reached around **\$43 billion USD** ^[4]

Quantum computing could generate **\$620 billion to \$1.27 trillion USD** in economic value by 2035 across industries ^[5]

Low-Code as a Key Enabler for making Quantum Computing Accessible

Cost, Complexity, and Talent Shortages Slow Quantum Adoption

Challenges



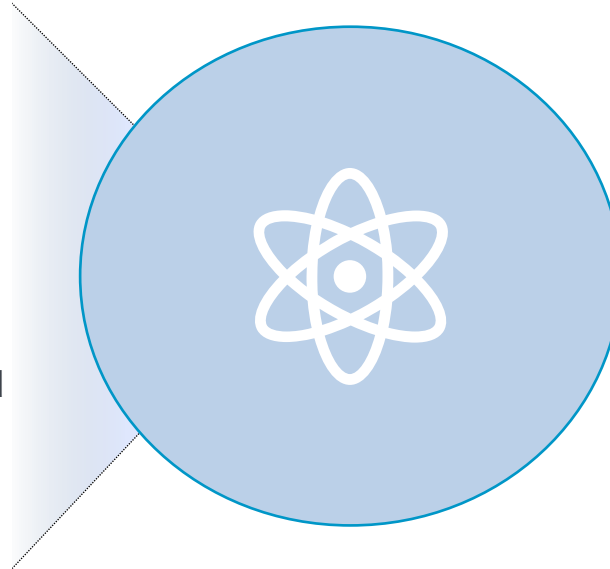
Developing and integrating quantum applications is costly, time-consuming, and highly complex



Shortage of quantum-skilled professionals

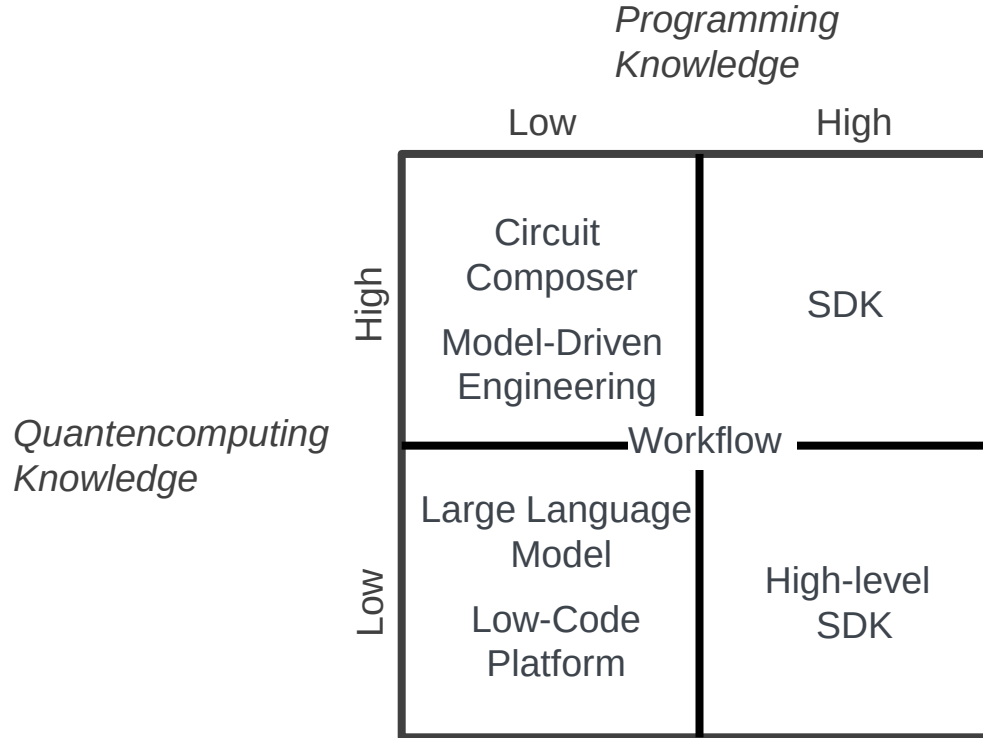


Fragmented quantum software landscape



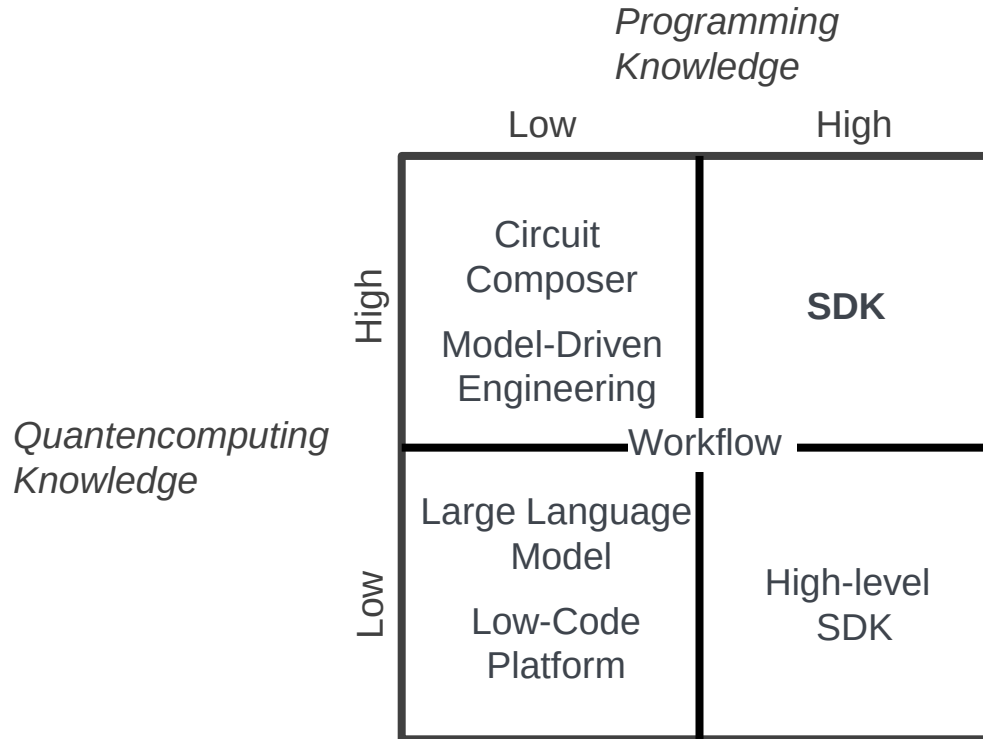
Current Quantum Software Landscape

Quantum software requires both programming and quantum knowledge



Current Quantum Software Landscape

Quantum software requires both programming and quantum knowledge



Low-Code as a Key Enabler for making Quantum Computing

Problem Statement

Challenges



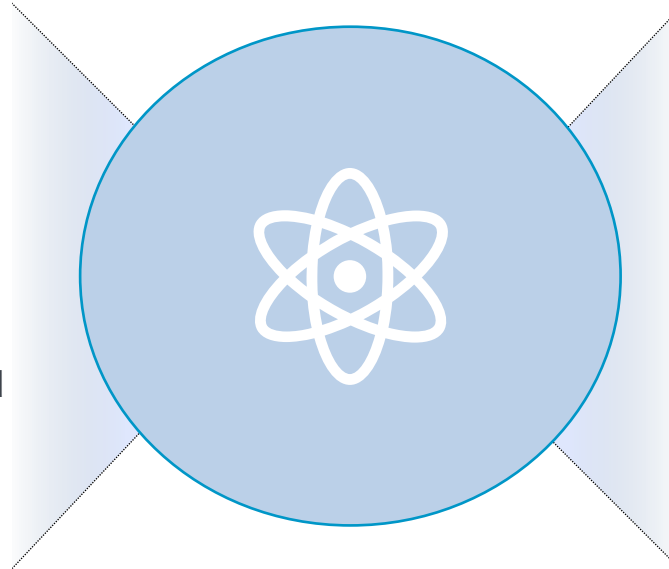
Developing and integrating quantum applications is costly, time-consuming, and highly complex



Shortage of quantum-skilled professionals



Fragmented quantum software landscape



Low-Code Solution



Code generation to enable the creation of quantum applications with little to no programming expertise



Use of visual and higher-level abstractions

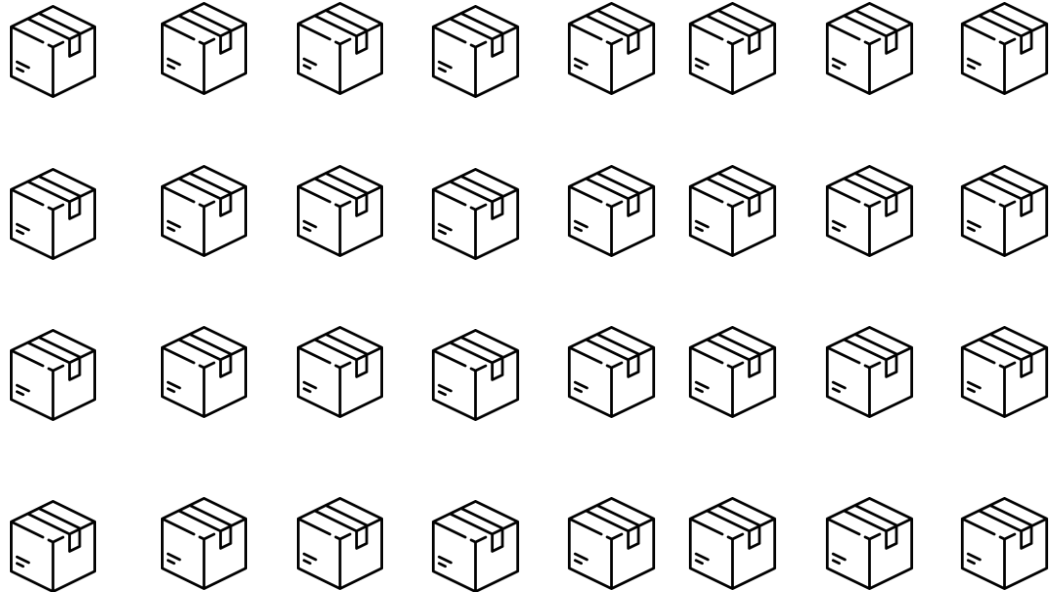
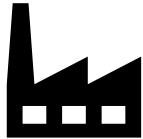


Integration of standards (e.g., OpenQASM, BPMN workflows)

Motivating Example

A Simple Business Question Hides Significant Quantum Complexity

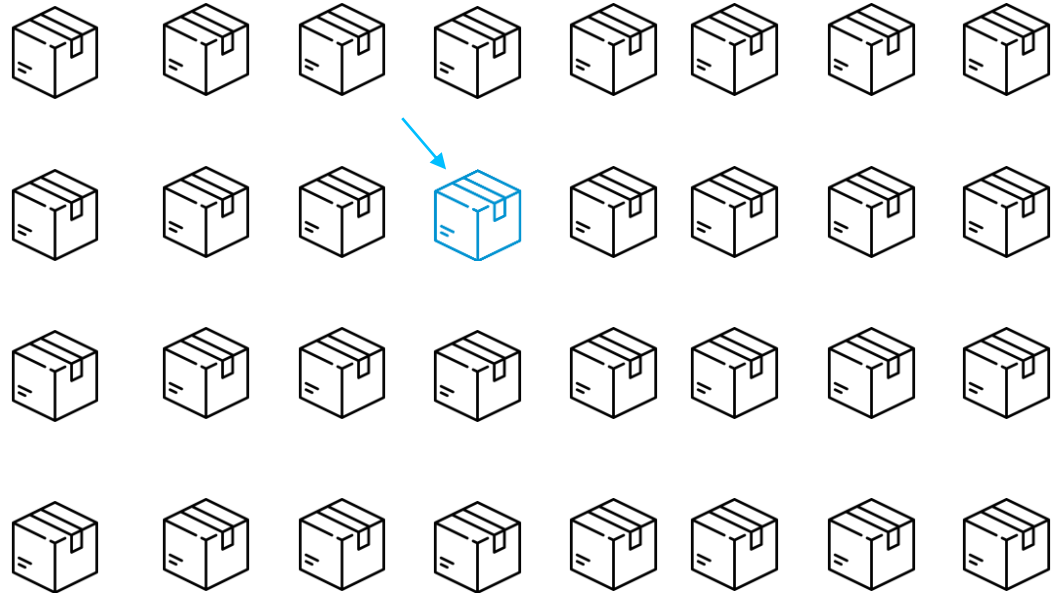
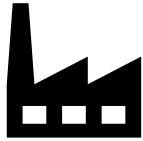
Find which shipment is critical.



Motivating Example

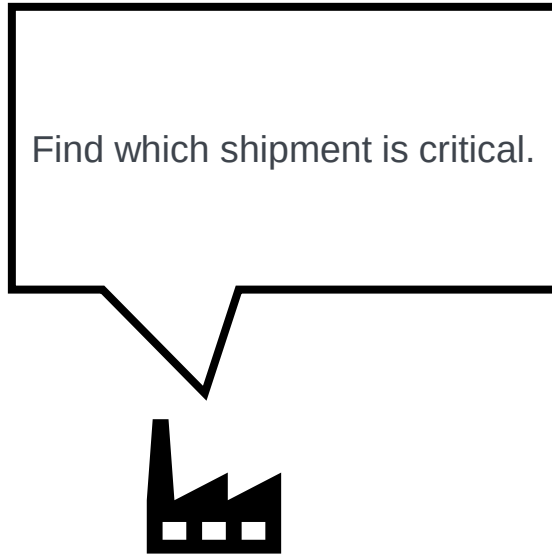
A Simple Business Question Hides Significant Quantum Complexity

Find which shipment is critical.



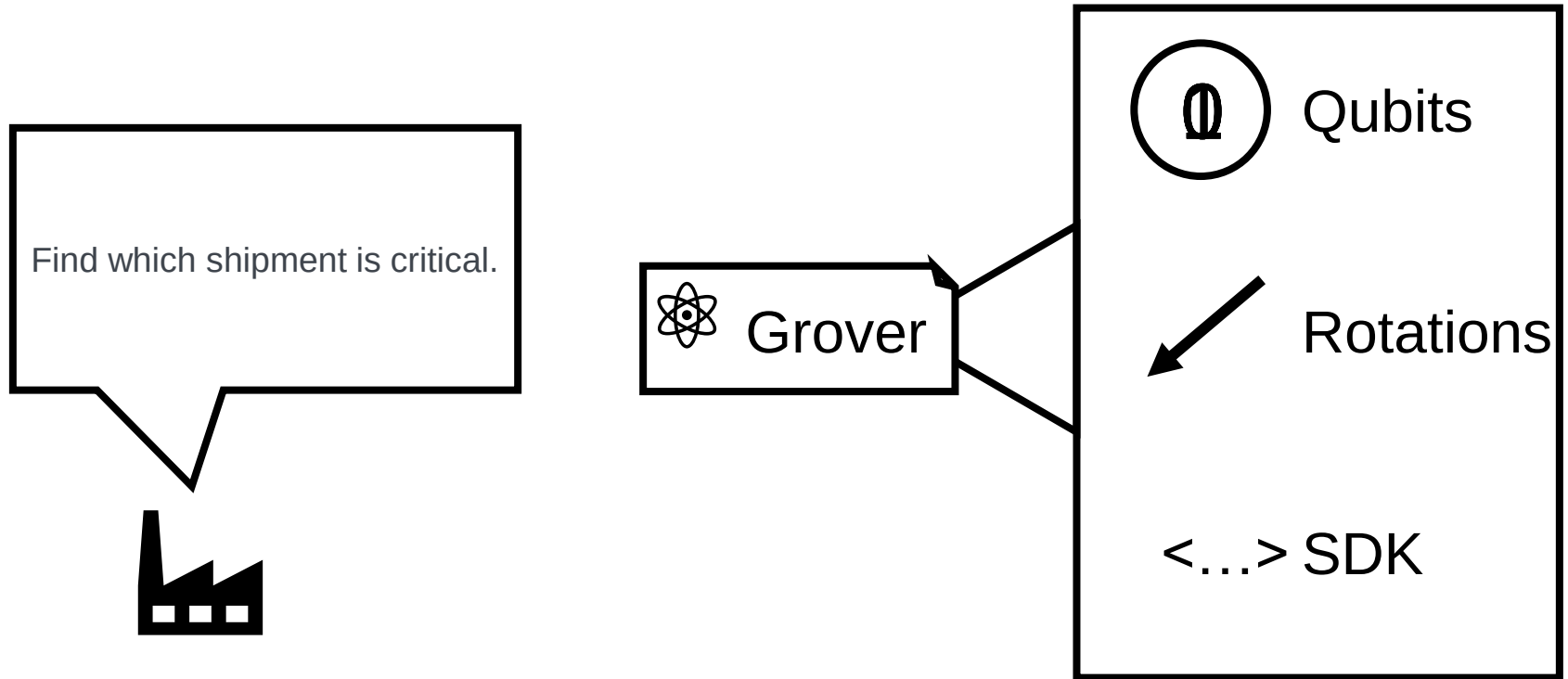
Motivating Example

Several Quantum Algorithms Could Answer the Same Question



Motivating Example

Implementing Even One Algorithm Demands Deep Technical Expertise



Problem: Steep Learning Curve

Qubits, Transformations, and SDKs Create a Steep Learning Curve



Steep Learning Curve

Qubits
Unitary Transformations
Software Development
Kits

Circuit Composer

Circuit Composers Lack Abstraction and Classical Code Support

- Circuit-level
- No possibility for classical code

The screenshot displays a quantum circuit designer interface with three main panels:

- Gates Panel:** A grid of quantum gates including H , T , $T^\dagger$, $\oplus$, $\otimes$, X , Y , Z , $\sqrt{X}$, $\sqrt{Z}$, R_X , R_Y , R_Z , P , and I .
- Quantum circuit Panel:** Shows a circuit with two qubits, $q[0]$ and $q[1]$. $q[0]$ has an H gate, followed by a CNOT gate with $q[1]$ as the target. $q[1]$ has a CNOT gate with $q[0]$ as the control. Below the circuit, there are labels $+$ and $-$ and a small table with columns 1 and 2.
- QASM code Panel:** Displays the following QASM code:

```
1 OPENQASM 3;  
2 include 'customgates.inc';  
3  
4 bit[2] c;  
5 qubit[2] q;  
6  
7 h q[0];  
8 cx q[0], q[1];  
9  
10 c[0] = measure q[0];  
11 c[1] = measure q[1];
```

[4] <https://thequantumlaend.de/quantum-circuit-designer>

Problem: Isolation & Fragmentation

Experts and Non-Experts Work in Isolation with No Shared Workspace



Steep Learning Curve

Qubits
Unitary Transformations
Software Development
Kits



**Experts & Non-Experts
work in Isolation**

One User
One Expertise Level
One View



**Growing teams have no
shared workspace**

Adaptations
Synchronizations
Track Changes

Solution: Experience-Aware Interfaces

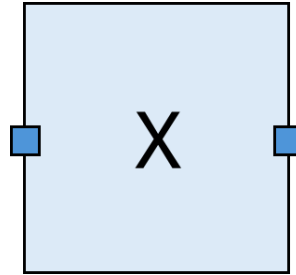
Interfaces That Adapt to the User's Expertise Level



**Experience-
aware
interfaces**

Circuit Primitives

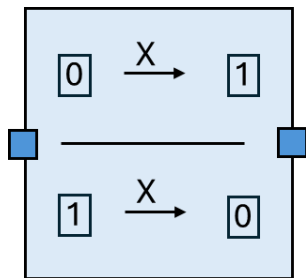
The Pioneer Variant Offers a Standard Gate Representation



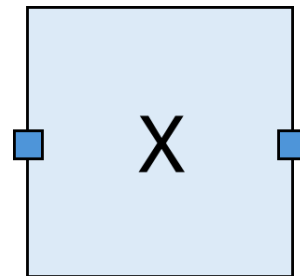
Pioneer

Circuit Primitives

Explorer and Pioneer Variants Support Different Expertise Levels



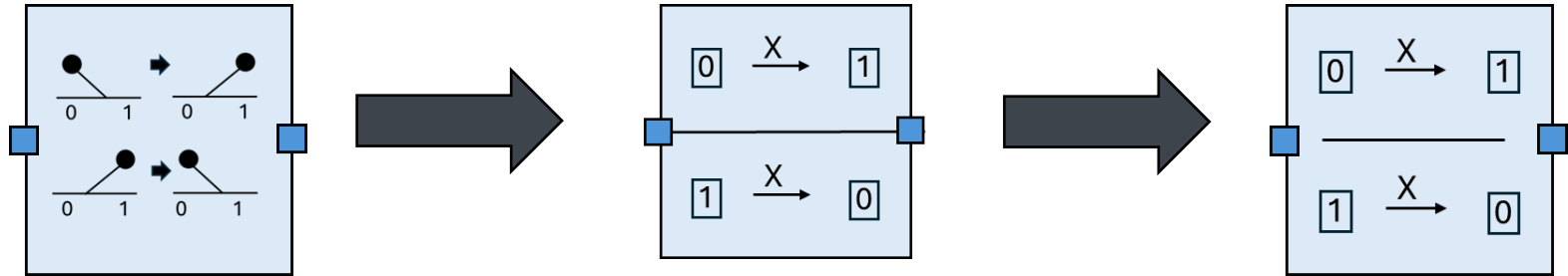
Explorer



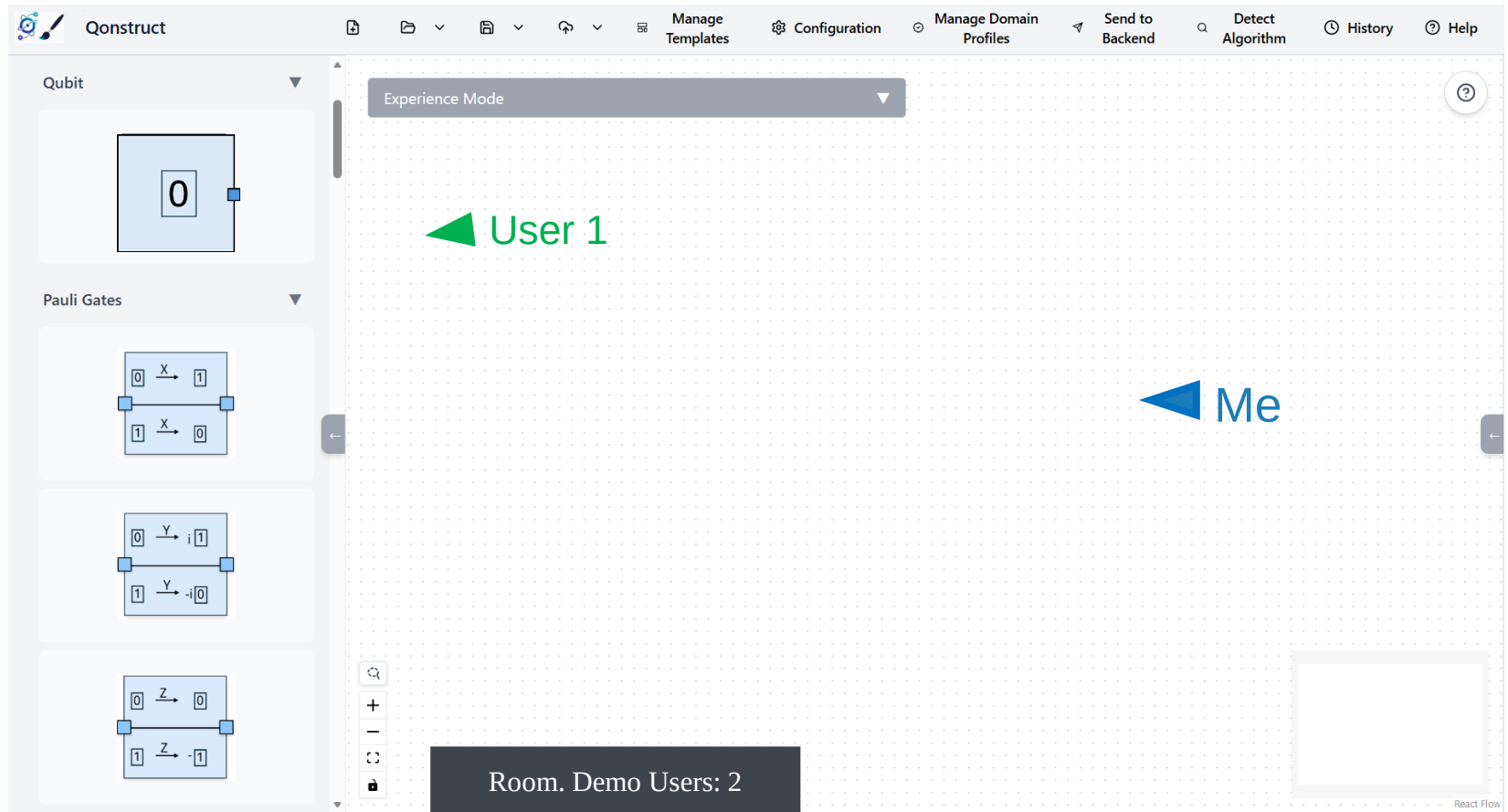
Pioneer

Circuit Primitives

Finding the right icon takes multiple iterations and evaluations



Synchronous Multi-User Editing



Solution: Synchronous Multi-User Editing

Real-Time Editing Lets Teams Collaborate Simultaneously



**Experience-
aware
interfaces**



**Synchronous
multi-user
editing**

Element-level locking

The screenshot shows the Qconstruct web application interface. On the left is a sidebar with a 'Qubit' section containing a single qubit block labeled '0' and a 'Pauli Gates' section containing three blocks: an X gate (0 to 1, 1 to 0), a Y gate (0 to i|1, 1 to -i|0), and a Z gate (0 to 0, 1 to -1). The main workspace is a grid with a top toolbar containing icons for 'Experience Mode', 'Manage Templates', 'Configuration', 'Manage Domain Profiles', 'Send to Backend', 'Detect Algorithm', 'History', and 'Help'. A green arrow labeled 'User 1' points to the top-left area of the grid. A blue arrow labeled 'Me' points to a qubit block labeled '0' in the center of the grid, which is highlighted with a blue glow. A dark grey box at the bottom left of the workspace displays the text 'Room. Demo Users: 2'. A small 'React Flow' logo is visible in the bottom right corner of the workspace.

Solution: Element-level locking



**Experience-
aware
interfaces**



**Synchronous
multi-user
editing**



**Element-level
locking**

Version History

Qubit

0

Pauli Gates

0

1

X

X

1

0

0

1

Y

Y

1

i

-i

1

0

0

1

Z

Z

0

-1

Experience Mode

User 1

Me

Room: Demo Users: 2

Collaboration History

User 1 added Qubit node

React Flow

Solution: Version History



**Experience-
aware
interfaces**



**Synchronous
multi-user
editing**



**Element-level
locking**



**Version
history**

Collaborative model validation

Qubit

0

Pauli Gates

0

X

1

1

X

0

0

Y

1

1

Y

-i0

0

Z

0

1

Z

-1

Experience Mode

User 1

Me

0

Encode Value

Encoding Type:

Basis Encoding

Bound:

Enter bound

Value

type: any

Output:

type: quantum circuit

Identifier

Size

Error:

Can not connect quantum output to classical input.

Room. Demo Users: 2

Collaboration History

Solution: Collaborative Model Validation



**Experience-
aware
interfaces**



**Synchronous
multi-user
editing**



**Element-level
locking**



**Version
history**



**Collaborative
model
validation**

Evaluation: Experience-Awareness

A Between-Subjects Study Tested Icon Comprehension

Participants

- 40 users in total
- Limited or no prior quantum computing experience

Study Design:

- Between-Subjects Design
 - Group 1: Icons in Quantum Platforms
 - Group 2: Icons of our platform
- Task: Functionality Description

Collected Metrics

- Correctness
- Task Completion Time
- Perceived ease of use (1–5)
- Clarity of concepts (1–5)
- Confidence in correctness (1–5)

Results – Experience-Awareness

New Icons Cut Task Time and Raised Confidence

<i>Metrics</i>	<i>Group 1 (general-purpose icons)</i>	<i>Group 2 (new icons)</i>
Correctness	1/20	20/20
Task Completion Time (per Icon)	60-120s	30s
Perceived ease of use	1/5	4/5
Clarity of concepts	1/5	4/5
Confidence in correctness	1/5	4/5

→ Group 1 led most often to misinterpretation, and the user hesitated with their answer

→ **New Icons reduced task time from minutes to seconds, raised confidence in correctness from 1 to 4**

Evaluation: Collaboration

Beginners, Experts, and Pairs Modeled Grover's Algorithm

Participants

- 40 users in total
- Limited or no prior quantum computing experience

Study Design:

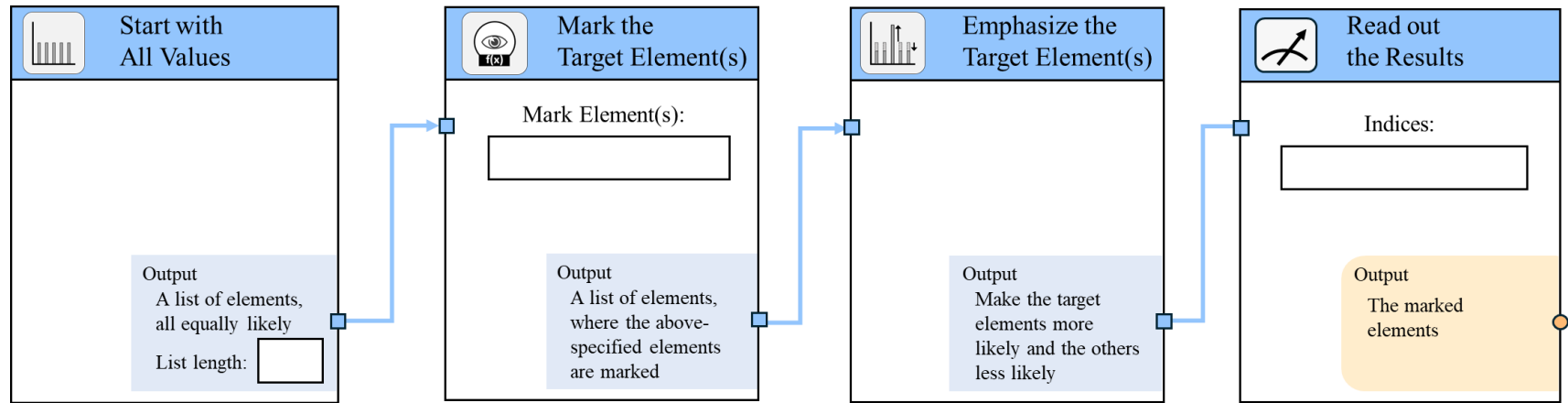
- Between-subjects design
 - Group 1 (5 beginners)
 - Group 2 (5 experts)
 - Group 3 (10 pairs)
- Task: Model Grover's search algorithm to find the element "2,7" in a list of 16 elements

Collected Metrics

- Correctness
- Task Completion Time
- Perceived ease of use (1–5)
- Clarity of concepts (1–5)
- Confidence in correctness (1–5)

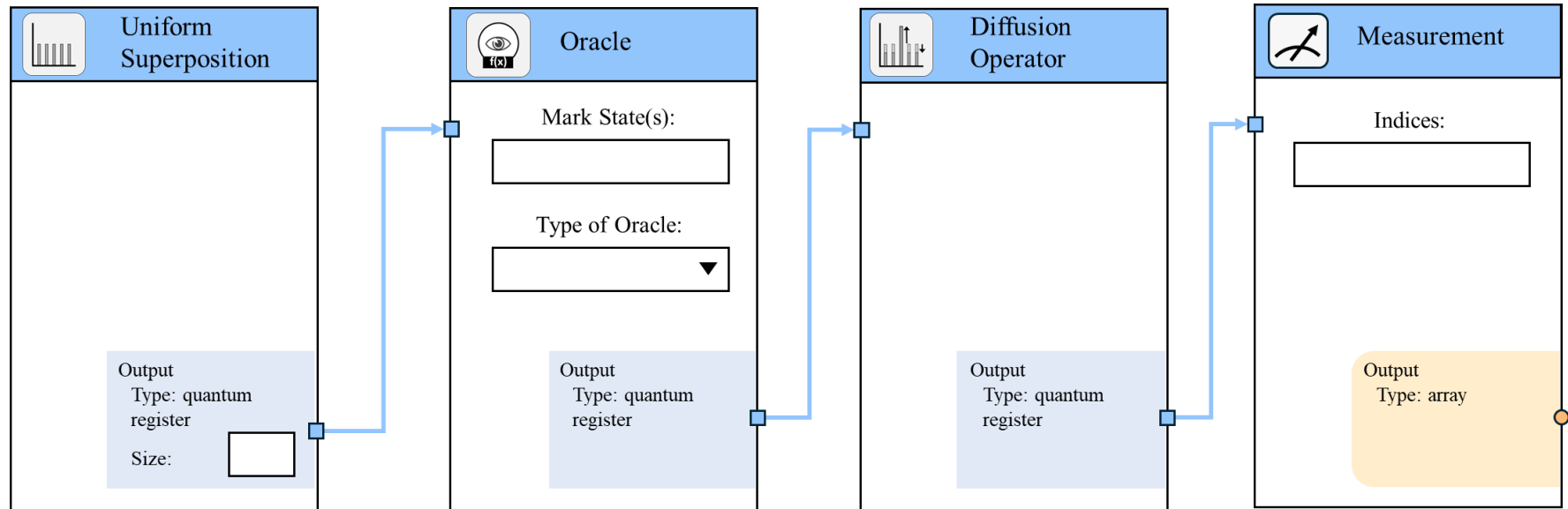
Evaluation – Beginner Group Grover

Beginners Modeled Grover's Algorithm at a High Level of Abstraction

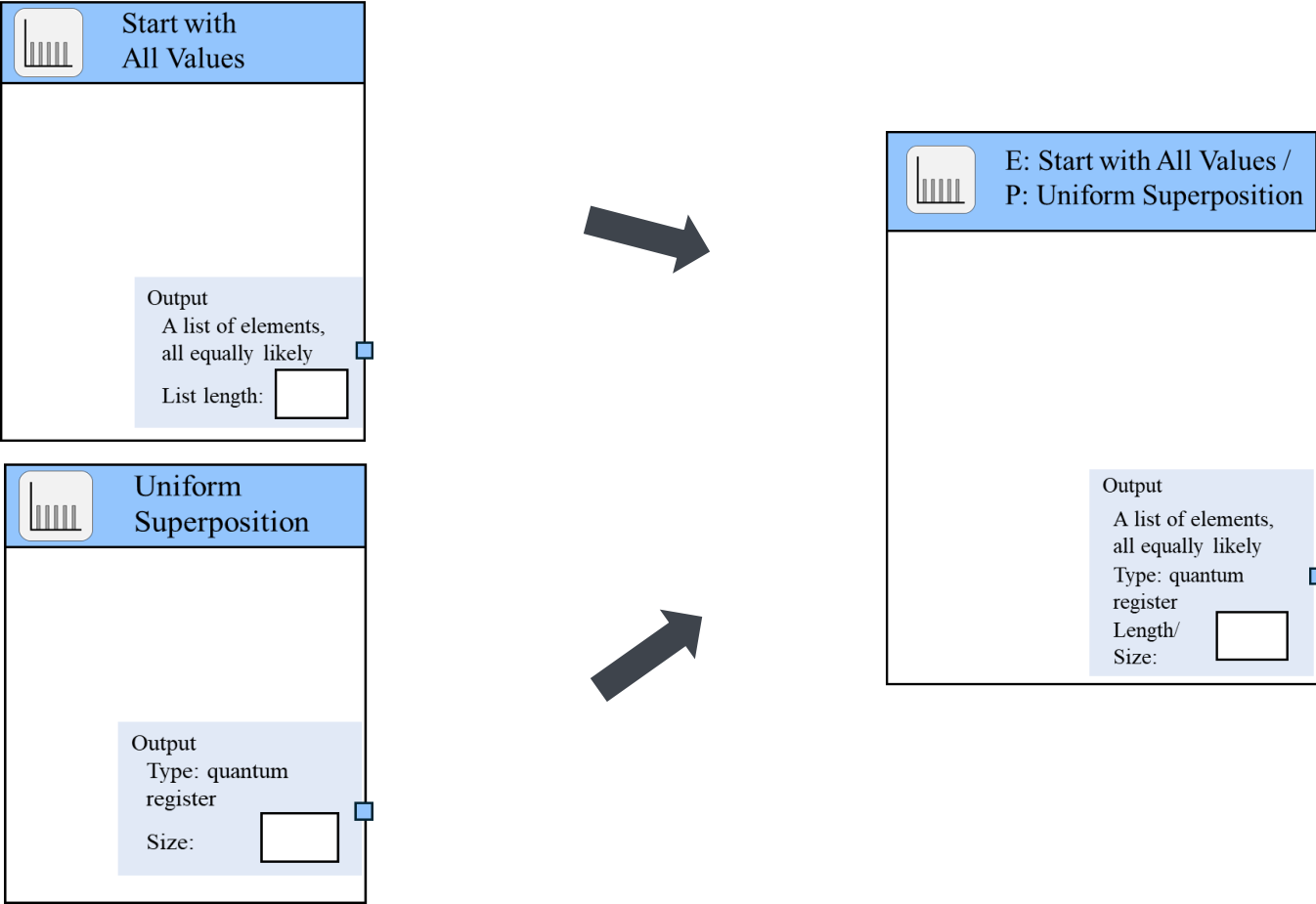


Evaluation – Expert Group Grover

Experts Modeled Grover's Algorithm with Full Technical Detail



Evaluation – Mixed Group Example



Results - Collaboration

Collaboration Cut Beginners' Time and Raised Their Confidence

<i>Metrics</i>	<i>Group 1 (beginner)</i>	<i>Group 2 (experts)</i>	<i>Group 3 (mixed group)</i>
Correctness	57%	100%	100%
Task Completion Time (min)	25	7	17
Perceived ease of use	3/5	5/5	5/5
Clarity of concepts	3/5	5/5	5/5
Confidence in correctness	3/5	5/5	5/5

→ Collaboration reduced task completion time for the beginners, raised correctness and confidence in correctness from 3 to 5

Stress Test

The Platform Stays Responsive with 40 Concurrent Users

<i>Metric</i>	<i>Observed Average</i>
Canvas Propagation Latency	24 ms ($\pm$ 4 ms)
In-Canvas Validator Error Trigger	48 ms ($\pm$ 2 ms)
Element Lock Acquisition Latency	14 ms ($\pm$ 2 ms)
Undo/Redo Global Canvas Synchronization	36 ms ($\pm$ 5 ms)

Limitations

Distributed Latency and Trust Assumptions Remain Untested

- Latency and synchronization overhead have not yet been geographically distributed benchmarked
- Assumes trusted collaborators

Conclusion

A Collaborative Low-Code Platform Reduces Quantum Fragmentation

- Introduced a collaborative quantum low-code platform to reduce fragmentation
- Combined:
 - Low-code modeling
 - Experience-awareness
 - Collaboration
- Aims to facilitate knowledge sharing and co-creation between experts and non-experts in quantum software development



Thank you for your attention 😊



**Thank you for
your attention 😊**

References

- [1] Nielsen, M. A. and Chuang, I. L. (2010). Quantum Computation and Quantum Information. Cambridge University Press.
- [2] Pérez-Castillo, R. and Piattini, M. (2022). Design of classical-quantum systems with UML. Computing, 104(11):2375–2403. Preskill, J. (2018). Quantum Computing in the NISQ era and beyond. Quantum, 2:79.
- [3] Havlíček, V., Córcoles, A. D., Temme, K., Harrow, A. W., Kandala, A., Chow, J. M., and Gambetta, J. M. (2019). Supervised learning with quantum-enhanced feature spaces. Nature, 567(7747):209–212.
- [4] <https://world-of-photonics.com/en/photonics-industry-portal/detail/quantum-technology-monitor-2024>
- [5] <https://www.bbva.com/en/innovation/how-the-world-map-of-quantum-computing-is-looking>